Day 4 – Network Cable Types and Tools

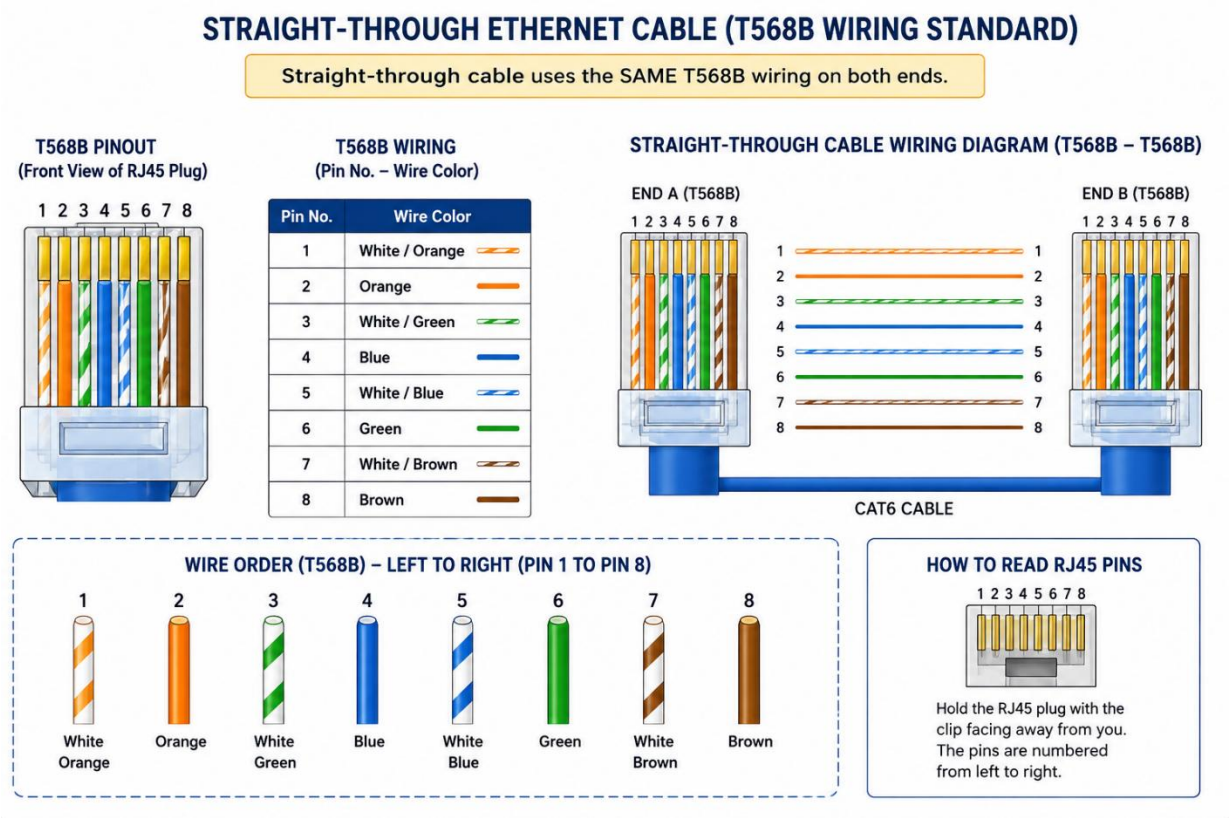
Task 1: Ethernet Cable vs Fiber Optic Cable

Point	Ethernet Cable	Fiber Optic Cable
1. Transmission	Uses electrical signals through copper wires.	Uses light signals through glass/plastic fibers.
2. Distance	Normally suitable for shorter distances, such as up to 100 m for standard twisted-pair Ethernet.	Suitable for very long distances, from hundreds of meters to many kilometers depending on the fiber and equipment.
3. Cost & Installation	Cheaper and easier to install and terminate using an RJ45 connector.	More expensive and requires specialized tools and skills.
Real-world example	Connecting PCs to a **Wi-Fi router or LAN switch** in a home, office, or gaming café.	Connecting buildings or switches in a **college campus or data-center backbone**.

Task 2: Straight-Through Ethernet Cable – T568B Wiring

A straight-through cable uses the **same T568B color arrangement on both ends.**

T568B Pin Order:



Task 3: Steps to Attach an RJ45 Connector to a CAT6 Cable

I referred to a YouTube demonstration showing CAT5e/CAT6 cable preparation, T568B wire arrangement, RJ45 insertion, crimping, and cable testing.

YouTube Video: How to Wire Up Ethernet Plugs – Switched On Network

Procedure – Attaching an RJ45 Connector to a CAT6 Cable

1. **Cut the CAT6 cable** to the required length using a cable cutter.
2. **Strip the outer jacket** about 2–3 cm from the end. Be careful not to damage the internal wires.
3. **Separate the four twisted pairs** and gently straighten all eight wires.
4. **Arrange the wires according to the T568B standard:**
    - Pin 1 – White/Orange
    - Pin 2 – Orange
    - Pin 3 – White/Green
    - Pin 4 – Blue
    - Pin 5 – White/Blue
    - Pin 6 – Green
    - Pin 7 – White/Brown
    - Pin 8 – Brown
5. **Straighten and trim the wires** evenly while keeping them in the correct order.

6. **Insert the wires into the RJ45 connector.** Make sure all eight wires reach the front of the connector and the cable jacket is properly inside.
7. **Check the wire order** carefully before crimping to avoid connection errors.
8. **Insert the RJ45 connector into the crimping tool** and squeeze firmly to secure the connector.
9. **Repeat the T568B arrangement on the other end** if making a straight-through Ethernet cable.
10. **Test the completed cable** using a LAN cable tester. The tester should show pins **1–8 in the correct sequence**.

Task 4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.

For a gaming café with **10 PCs**, I would choose **CAT6 Ethernet cables** because they provide high speed, low latency, and reliable connections for LAN gaming. The cable runs inside a café are usually short, so fiber optics would be unnecessary and more expensive. I would use an **RJ45 crimper** to attach RJ45 connectors to the CAT6 cables because it is simple, affordable, and easy to use. Overall, **CAT6 Ethernet + RJ45 crimper** is the best choice for speed, distance, and budget.

